

The Power-Character and Bounded-Support Stability Theorem

Proof, scope, and exact matrix verification

Lambda Distributions workspace

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Abstract

For an arbitrary finite group and honest complex representation we derive the sigma-MGF from character values on all powers. For permutation actions this becomes an exact fixed-point and cycle formula; for coset actions it becomes a centralizer–subgroup–intersection formula, and for conjugation it becomes a centralizer formula. We then prove coefficientwise fixed-field stability for bounded flags in the classical growing-rank towers. A marimo notebook checks the linear, parabolic, conjugation, and stability statements by explicit matrices or generator orbits. The proof is general; finite computations are reported only as independent exact verification.

Status. Equations (1)–(9) and the type- A threshold (11) are exact theorems. The formed-space statement is coefficientwise stability with a finite tower-dependent threshold, not a claimed universal numerical threshold. Complete flags are excluded: their degree-two coefficient already grows with the Weyl group.

1 Common target and universal character-power formula

Let G be finite and let $\rho : G \rightarrow GL(V)$ be an honest finite-dimensional complex representation with character χ_V . For a partition $\tau = (\tau_1, \dots, \tau_s)$ set

$$W_\tau(V) = \bigotimes_{j=1}^s \text{Sym}^{\tau_j} V, \quad |\tau| = \sum_j \tau_j,$$

and define

$$\mathcal{S}_{G,V} = \mathbb{E}_{g \in G} [\text{Exp}_\sigma(X_V(g)h_1)] = \sum_\tau \dim W_\tau(V)^G m_\tau.$$

Theorem 1 (Universal power-character formula). *One has the exact identities*

$$\mathcal{S}_{G,V} = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \det(I - t_i \rho(g))^{-1}, \tag{1}$$

$$= \frac{1}{|G|} \sum_{g \in G} \exp \left(\sum_{r \geq 1} \frac{\chi_V(g^r)}{r} p_r \right), \tag{2}$$

$$= \sum_{[g] \subseteq G} \frac{1}{|C_G(g)|} \exp \left(\sum_{r \geq 1} \frac{\chi_V(g^r)}{r} p_r \right). \tag{3}$$

Thus the series is determined by conjugacy classes, class power maps, and the character values. Its coefficient is

$$[m_\tau]\mathcal{S}_{G,V} = \frac{1}{|G|} \sum_{g \in G} \prod_{j=1}^s \left(\sum_{\lambda \vdash \tau_j} \frac{1}{z_\lambda} \prod_{r \geq 1} \chi_V(g^r)^{m_r(\lambda)} \right). \quad (4)$$

Proof. On $W_\tau(V)$ the Reynolds operator $|G|^{-1} \sum_g \rho_{W_\tau}(g)$ is an idempotent with image $W_\tau(V)^G$. Its trace is therefore the invariant dimension. Characters multiply on tensor products, while

$$\sum_{a \geq 0} \text{Tr}(\text{Sym}^a A) t^a = \det(I - tA)^{-1}.$$

Multiplying one copy for each variable and averaging proves (1). The formal identity

$$-\log \det(I - tA) = \sum_{r \geq 1} \frac{\text{Tr}(A^r)}{r} t^r$$

and $\text{Tr}(\rho(g)^r) = \chi_V(g^r)$ prove (2); grouping elements by class proves (3). Finally $h_a = \sum_{\lambda \vdash a} p_\lambda / z_\lambda$ gives (4). $\square$

Remark 2 (Character specializations). *For a genuine Steinberg or a specified genuine Weil representation, insert its character into (3). For a Deligne–Lusztig virtual character $R_T^G(\theta)$ the same expression is interpreted in the completed Grothendieck lambda-ring; its coefficients are virtual multiplicities and need not be nonnegative. These are specializations of the theorem, not claims that all such character values collapse to one uniform product.*

2 Permutation and parabolic quotient groups

Suppose G acts on a finite set X , and use the honest complex permutation module $V = \mathbb{C}[X]$. Let $c_r(g; X)$ count the r -cycles and put $F_d(g; X) = |\text{Fix}_X(g^d)|$.

Proposition 3 (Permutation-cycle formula). *For every finite action,*

$$c_r(g; X) = \frac{1}{r} \sum_{d|r} \mu(r/d) F_d(g; X), \quad (5)$$

$$\mathcal{S}_{G,X} = \frac{1}{|G|} \sum_{g \in G} \prod_{i, r \geq 1} (1 - t_i^r)^{-c_r(g; X)}, \quad (6)$$

$$[m_\tau]\mathcal{S}_{G,X} = \# \left(G \backslash \prod_j \text{Sym}^{\tau_j} X \right). \quad (7)$$

Proof. An r -cycle has determinant factor $1 - t^r$ and contributes all of its r points to F_d precisely when $r \mid d$. Hence $F_d = \sum_{r|d} r c_r$, and Möbius inversion gives (5). The determinant factors give (6). The monomial basis of $W_\tau(\mathbb{C}[X])$ is the displayed product of multiset spaces, so its invariant dimension is the number of orbits, proving (7). $\square$

Proposition 4 (Coset and parabolic cycles). *For $X = G/H$,*

$$F_d(g; G/H) = \frac{|C_G(g^d)|}{|H|} |(g^d)^G \cap H|, \quad (8)$$

$$c_r(g; G/H) = \frac{1}{r} \sum_{d|r} \mu(r/d) \frac{|C_G(g^d)|}{|H|} |(g^d)^G \cap H|. \quad (9)$$

Substitution in (6) gives the complete exact sigma-MGF. In particular this applies to $G = \mathbf{G}^F$ and every F -stable parabolic P , including projective points, Grassmannians, polar Grassmannians, and fixed-shape partial flags.

Proof. The coset xH is fixed by g^d exactly when $x^{-1}g^d x \in H$. Each element of $(g^d)^G \cap H$ has $|C_G(g^d)|$ conjugators x . Divide by $|H|$ for the number of representatives of a coset and then apply (5). $\square$

3 Conjugation groups

For conjugation on G itself,

$$F_d(g; G_{\text{conj}}) = |C_G(g^d)|.$$

Therefore

$$c_r^{\text{conj}}(g) = \frac{1}{r} \sum_{d|r} \mu(r/d) |C_G(g^d)|, \quad \mathcal{S}_{G, \text{conj}} = \frac{1}{|G|} \sum_g \prod_{i,r} (1 - t_i^r)^{-c_r^{\text{conj}}(g)}. \quad (10)$$

On a fixed conjugacy class C , replace the fixed-point count by $|C \cap C_G(g^d)|$. Hence the conjugation series is determined by centralizers, class intersections, and power maps.

4 Fixed-field stability for bounded flags

Fix q and a flag shape $\mathbf{d} = (d_1 < \dots < d_s)$, with $d = d_s$ independent of ambient rank. Let $X_n = \text{Flag}_{\mathbf{d}}(\mathbb{F}_q^n)$.

Theorem 5 (Type-A bounded-support stability). *For every partition τ of total degree D ,*

$$[m_\tau] \mathcal{S}_{GL_n(q), X_n} \quad \text{is independent of } n \text{ for } n \geq dD. \quad (11)$$

The same holds for $PGL_n(q)$. For $SL_n(q)$ and $PSL_n(q)$ the safe range is $n > dD$. The stable coefficient is

$$a_{\tau, \mathbf{d}}(q) = \sum_{r=0}^{dD} \# \left(GL_r(q) \setminus \mathcal{C}_{\tau, \mathbf{d}}^{\text{span}}(r) \right), \quad (12)$$

where the configuration space consists of τ -colored flag multisets whose top subspaces span $\mathbb{F}_q^r$.

Proof. A degree- D configuration uses D flags with multiplicity. The sum of their top subspaces, called the support, has dimension at most dD . Two configurations are $GL_n(q)$ -conjugate exactly when their supported labelled configurations are linearly isomorphic, because every isomorphism between the supports extends across complements. Once $n \geq dD$, every possible support already embeds, so no new orbit type can appear. Sorting by the support dimension gives (12). Scalar matrices act trivially, so the PGL orbits agree. If $n > dD$, a nonzero complementary direction can correct the determinant of any extension while fixing the target support; this gives the stated SL and PSL range. $\square$

Proposition 6 (Formed-space towers). *Fix one symplectic, unitary, odd-orthogonal, or one of the two even-orthogonal towers, and let X_n be the totally isotropic or singular flags of fixed shape. For every τ there is a finite threshold $N_{\mathcal{T}}(\mathbf{d}, \tau)$ after which $[m_{\tau}]S_{G_n, X_n}$ is constant.*

Proof. The span of a degree- D configuration again has bounded dimension. Its orbit is determined by the labelled flags together with the restricted Hermitian, alternating, or quadratic form and the multiplicities. In characteristic two the quadratic form, not merely its polar form, is retained. There are only finitely many such bounded-dimensional supported types over a fixed field. Witt extension embeds and identifies each type once the chosen tower is large enough; taking the maximum of these finitely many embedding ranks gives the threshold. Determinant or spinor-norm correction may require enlarging the unused nondegenerate complement before passing to special, derived, or projective groups. $\square$

Remark 7 (Why complete flags are excluded). *For G/B , Bruhat decomposition gives $[m_{(1,1)}]S_{G, G/B} = |B \backslash G/B| = |W|$. This grows with rank, so bounded shape is an essential hypothesis.*

5 Representation scope for finite Lie-type groups

Projective, subspace, flag, and conjugacy actions are honest complex permutation representations. Formula (3) also applies to every specified genuine complex character, including Steinberg and chosen Weil representations. The defining-characteristic action on $\mathfrak{g}(\mathbb{F}_q)$ requires care:

- as a finite set with conjugation action it is an honest permutation representation, with $F_d(g) = q^{\dim_{\mathbb{F}_q} C_{\mathfrak{g}}(g^d)}$;
- as a linear module over $\mathbb{F}_q$ it is not automatically a complex representation, so a cross-characteristic realization or lift must be specified before applying (2).

6 Group-separated exact verification

The executable companion is `proved_matrix_groups/packages/power_character_bounded_flag_stability/notebook.py`. It calls a plain Python verification module, so the same checks run without a notebook server.

6.1 Linear group: $S_3 \cong GL_2(2)$ and its Steinberg module

Every one of the six integral 2×2 matrices of the standard S_3 representation is constructed. For each τ , the program builds the matrices of every $\text{Sym}^{\tau_j} V$, takes their Kronecker product, averages the actual matrices, and computes the Reynolds rank. Independently it uses only $\chi(g^r)$ and Newton's recurrence to evaluate (4). In the order

$$(1), (2), (1, 1), (3), (2, 1), (1, 1, 1),$$

both routes give

$$(0, 1, 1, 1, 1, 1).$$

The largest projector-idempotence error is 5.56×10^{-17} .

6.2 Parabolic group: $GL_3(2)$ on G/P

The program enumerates all 168 invertible 3×3 matrices over $\mathbb{F}_2$ and their action on the seven projective points. It separately computes all six conjugacy classes, the point stabilizer P of order 24, centralizers, and class intersections with P . Formula (8) reproduces every needed fixed-point count and reconstructs the cycles of all 168 elements. Actual permutation-matrix averages, cycle factors, and direct multiset-orbit traversal agree on

$$(1, 2, 2, 4, 5, 6).$$

There are 715 direct fixed-point equalities and 168 full cycle reconstructions.

6.3 Conjugation group: S_3 on itself

The six conjugation permutations are built from the group law. Direct fixed points agree with $|C_G(g^d)|$ for every required power, and all six cycle decompositions are reconstructed. Matrix averages, (10), and direct orbits agree on

$$(3, 8, 11, 17, 32, 49).$$

6.4 Stable type-A groups

Transvections generate the actions without enumerating the large ambient groups. Direct configuration-orbit traversal gives:

τ for projective points	$n = 1$	$n = 2$	$n = 3$	$n = 4$
(1)	1	1	1	1
(2)	1	2	2	2
(1, 1)	1	2	2	2
(3)	1	3	4	4
(2, 1)	1	4	5	5
(1, 1, 1)	1	5	6	6

The onset is exactly consistent with $n \geq |\tau|$ when $d = 1$. For $GL_n(2)$ on $\text{Gr}_2(n)$, both $\tau = (2)$ and $(1, 1)$ give three orbits at $n = 4$ and again at $n = 5$, matching the threshold $n \geq 2|\tau| = 4$ and the intersection-dimension classification.

Verification result. All 26 reported coefficient or stability rows pass. These computations test the formulas at small representative dimensions with distinct failure modes; the all-rank claims rest on the proofs above.

7 Reproduction

From the repository root, run

```
python3 -m for_this_guy.power_character_bounded_flag_stability.verification
.venv/bin/marimo run for_this_guy/\
    power_character_bounded_flag_stability/notebook.py
```

References

- [1] S. Howe, *Random matrix statistics and zeroes of L -functions via probability in lambda-rings*, [arXiv:2412.19295](#).
- [2] J. Huang, *Some extensions of Witt's theorem*, [arXiv:0705.3839](#).
- [3] T. Thomas, *The character of the Weil representation*, [arXiv:math/0610644](#).
- [4] P. Magyar, J. Weyman, and A. Zelevinsky, *Multiple flag varieties of finite type*, [arXiv:math/9805067](#).